

SHADOWFOX

DATA ANALYST INTERNSHIP

INTERNSHIP REPORT

BEGINNER LEVEL

Project Title

**Superstore Sales Analysis and
Business Performance Dashboard**

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Internship Role: Data Analyst Intern

Organization: ShadowFox

GitHub Repository : https://github.com/prensi01/ShadowFox_Data_Analyst_Beginner

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1. Introduction

As part of the ShadowFox Data Analyst Internship, I completed a Beginner-level sales analysis project using a Superstore sales dataset.

The objective of this project was to clean and organize the dataset, analyze sales and profit performance, identify important business trends, and present the findings through charts and a dashboard.

The analysis focuses on overall sales, profit, quantity, orders, category performance, monthly sales trends, and product-level performance.

2. Project Objective

The main objectives of this project were:

- To clean and organize the sales dataset.
 - To identify and handle duplicate records.
 - To check important columns for missing values.
 - To analyze total sales, profit, quantity, and orders.
 - To understand monthly sales trends.
 - To compare sales and profit across product categories.
 - To identify high-performing products.
 - To identify products with negative profit.
 - To calculate profit margins.
 - To create meaningful charts and a simple business dashboard.
 - To provide business-oriented insights and recommendations.
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3. Dataset Description

The project uses a **Superstore Sales Dataset** containing sales and customer-related information.

The dataset contains **10,194 data records** and **21 columns**.

The major columns used in the analysis include:

- Row ID
- Order ID
- Order Date
- Ship Date

- Ship Mode
- Customer ID
- Customer Name
- Segment
- Country/Region
- City
- State/Province
- Postal Code
- Region
- Product ID
- Category
- Sub-Category
- Product Name
- Sales
- Quantity
- Discount
- Profit

The dataset contains information related to orders, customers, products, sales, quantity, discount, and profit.

4. Tools and Technologies Used

The following tools were used for this project:

- **Microsoft Excel** – Data cleaning, analysis, PivotTables, charts, and dashboard creation.
 - **GitHub** – Project submission and version management.
 - **Superstore Dataset** – Source data for the analysis.
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5. Data Cleaning and Preparation

Before performing the analysis, the dataset was cleaned and prepared in Microsoft Excel.

The following steps were performed:

5.1 Data Formatting

The dataset was converted into an Excel Table to make filtering, sorting, and analysis easier.

5.2 Duplicate Check

The dataset was checked for duplicate records.

Result: No duplicate values were found during the duplicate check.

5.3 Missing Value Check

Important columns such as:

- Order Date
- Customer ID
- Customer Name
- Region
- Category
- Product Name
- Sales
- Quantity
- Profit

were checked for blank values.

Result: No missing values were found in these important fields.

5.4 Date Cleaning

The Order Date column contained dates in different formats. A separate **Clean Order Date** column was created to standardize the date values.

This cleaned date column was then used for monthly and yearly sales analysis.

5.5 Data Validation

The cleaned dataset was reviewed to ensure that the important fields were usable for analysis and dashboard creation.

6. Data Analysis

After cleaning the dataset, different analyses were performed using Excel PivotTables.

The analysis included:

- Overall business performance
- Monthly sales trends
- Category-wise sales and profit
- Product-level sales performance
- Profit margin analysis
- Loss-making product identification

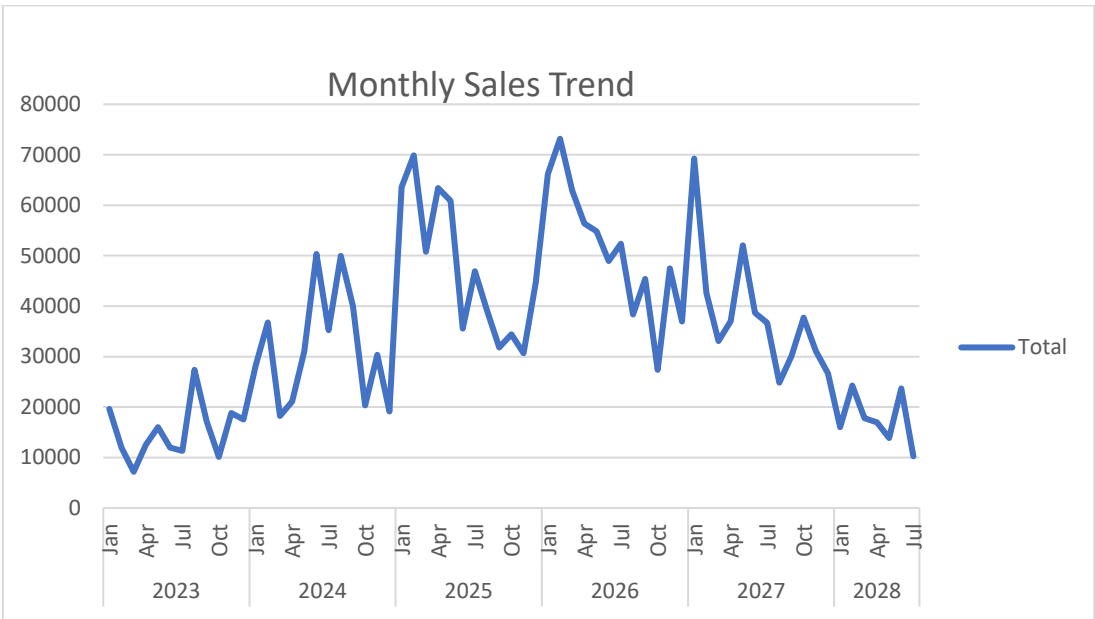


Figure 2: Monthly Sales Trend

7. Key Performance Indicators

The major KPIs calculated from the dataset are:

KPI	Value
Total Sales	₹2,326,534.35
Total Profit	₹292,296.81
Total Quantity	38,654
Total Orders	5,111
Overall Profit Margin	12.56%

These KPIs provide a high-level overview of the business performance.

8. Dashboard and Visualizations

An Excel dashboard was created to present the major findings in an easy-to-understand format.

The dashboard includes KPI cards for:

- Total Sales
- Total Profit
- Total Quantity
- Total Orders

It also includes visualizations for:

- Monthly Sales Trend
- Category Performance
- Top Product Performance

The dashboard helps users quickly understand overall business performance without going through the complete dataset.

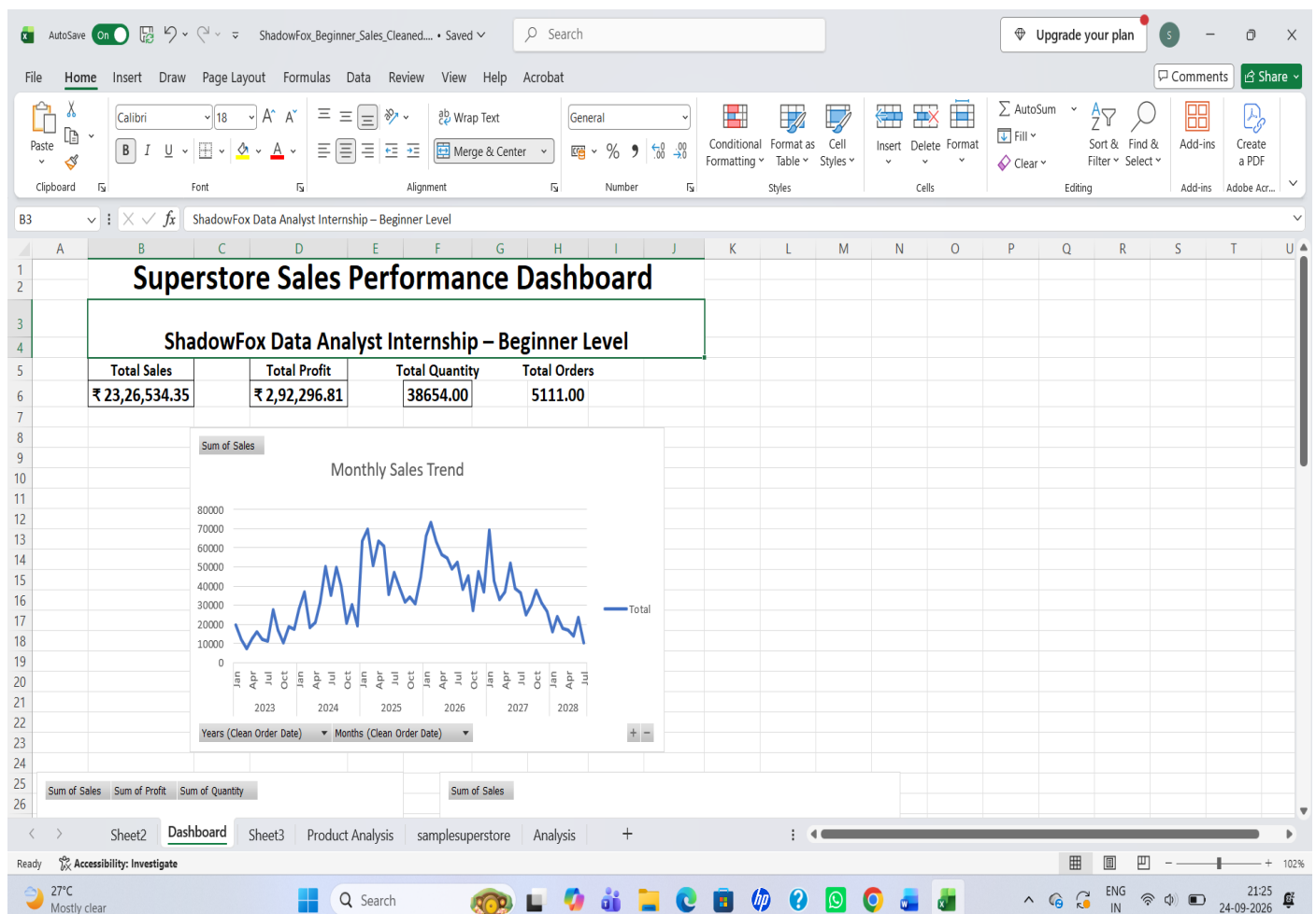


Figure 1: Superstore Sales Performance Dashboard – KPIs and Monthly Sales Trend

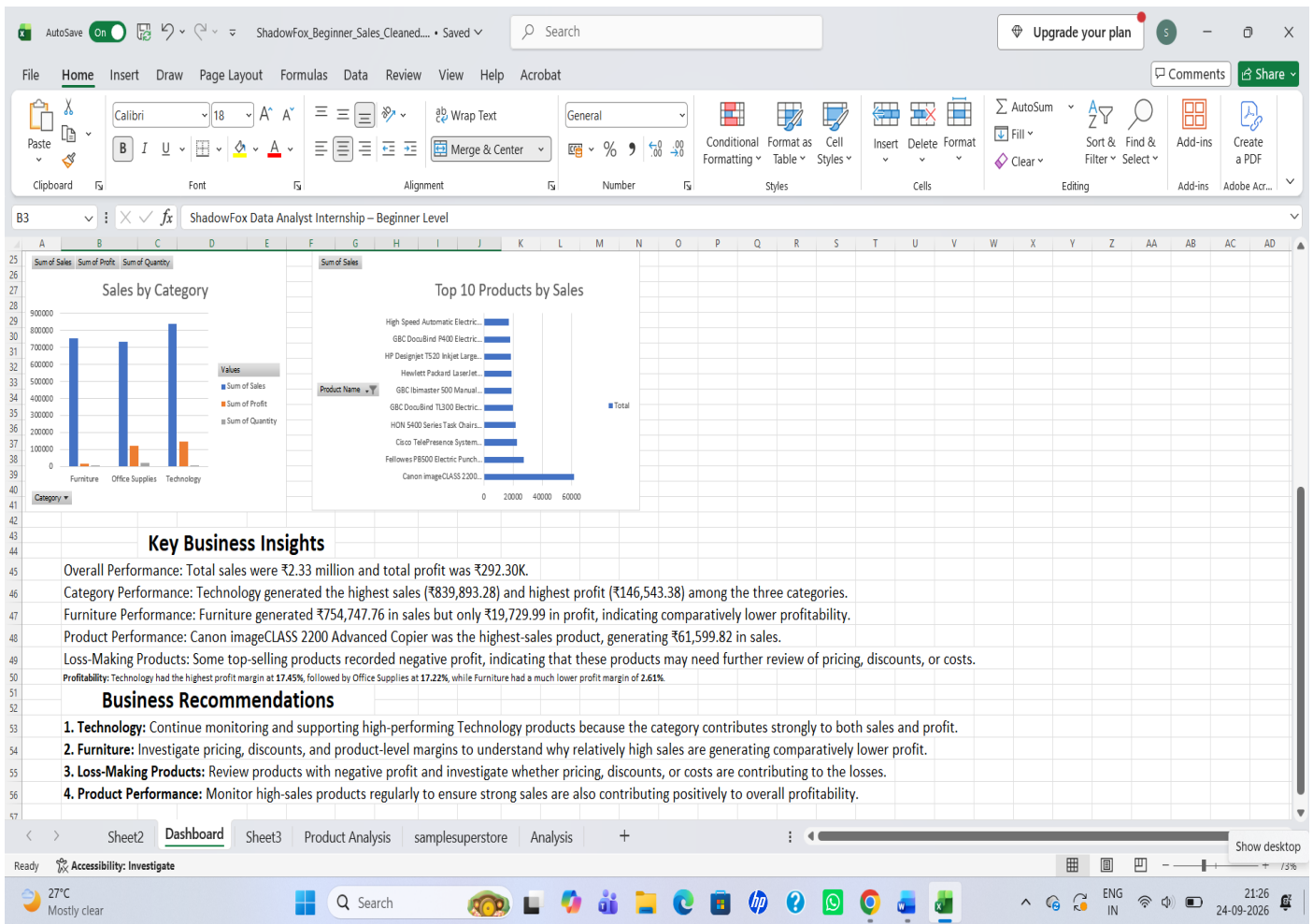


Figure 5: Superstore Sales Dashboard – Category, Product Analysis and Business Insights

9. Key Business Insights

9.1 Overall Performance

The business generated total sales of approximately **₹2.33 million** and total profit of approximately **₹292.30K**.

9.2 Category Performance

The **Technology** category generated the highest sales of approximately **₹839,893.28** and the highest profit of approximately **₹146,543.38**.

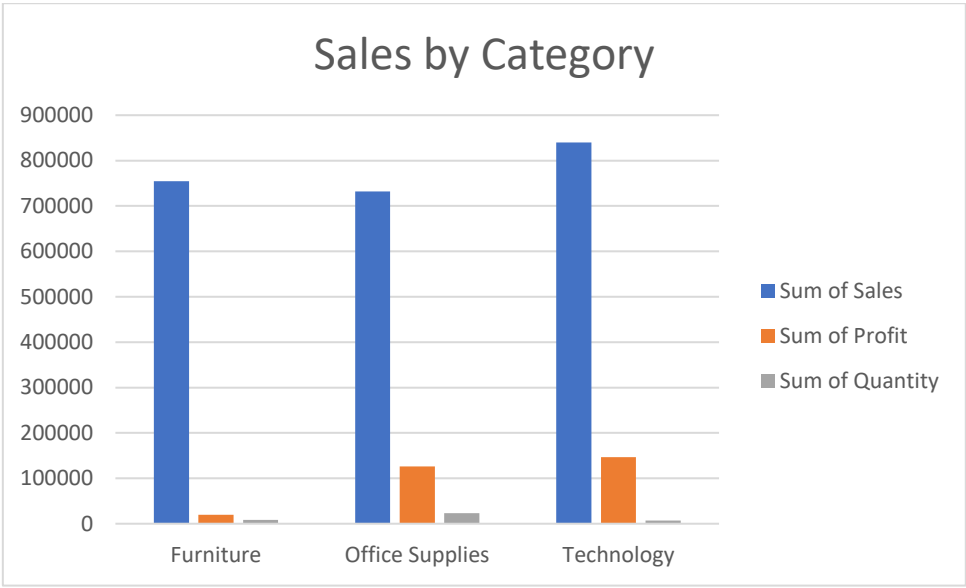


Figure 3: Category-wise Sales and Profit

9.3 Furniture Performance

Furniture generated approximately **₹754,747.76** in sales but only **₹19,729.99** in profit.

Its profit margin was approximately **2.61%**, which is considerably lower than the other two categories.

9.4 Office Supplies Performance

Office Supplies generated approximately **₹731,893.31** in sales and **₹126,023.44** in profit.

Its profit margin was approximately **17.22%**.

9.5 Technology Profitability

Technology had a profit margin of approximately **17.45%**, making it the category with the highest calculated profit margin in this analysis.

9.6 Top Product

The **Canon imageCLASS 2200 Advanced Copier** was the highest-sales product in the selected top-product analysis, generating approximately **₹61,599.82** in sales.

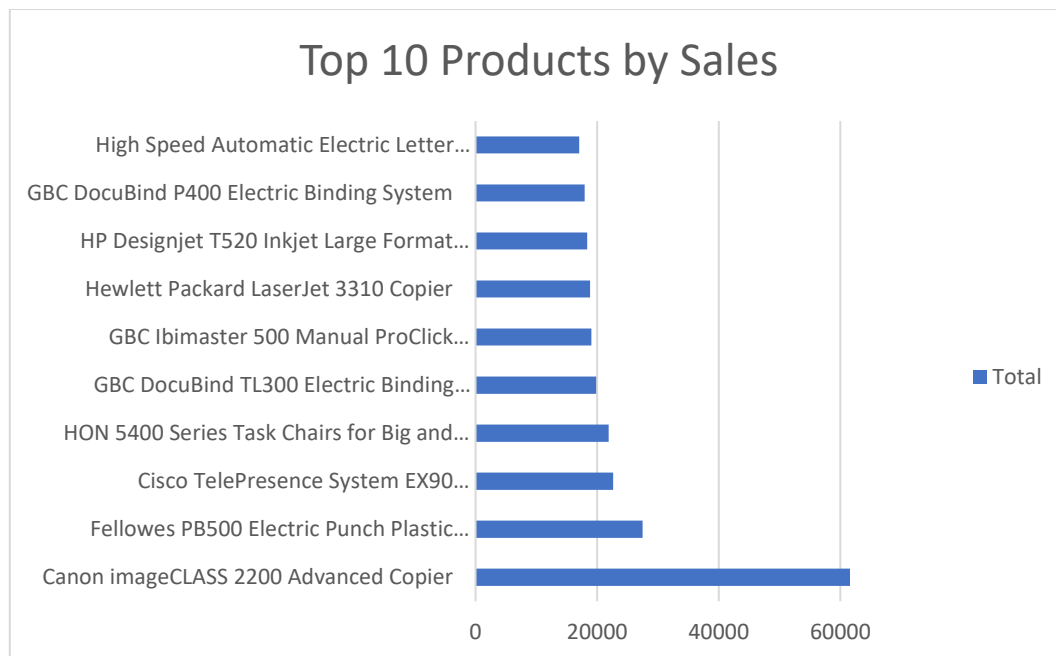


Figure 4: Top 10 Products by Sales

9.7 Loss-Making Products

Some products with relatively high sales recorded negative profit.

For example:

- Cisco TelePresence System EX90 Videoconferencing Unit: approximately **-₹1,811.08 profit**
- GBC DocuBind P400 Electric Binding System: approximately **-₹1,878.17 profit**
- High Speed Automatic Electric Letter Opener: approximately **-₹262.00 profit**

This indicates that sales volume alone does not always result in positive profitability.

10. Business Recommendations

Based on the analysis, the following recommendations can be considered:

10.1 Monitor Technology Products

Technology is contributing strongly to both sales and profit. High-performing Technology products should be monitored regularly to understand the factors contributing to their performance.

10.2 Investigate Furniture Profitability

Furniture has relatively high sales but a much lower profit margin.

Pricing, discounts, product costs, and individual product margins should therefore be reviewed.

10.3 Review Loss-Making Products

Products generating negative profit should be analyzed further.

The business can investigate:

- Pricing
- Discounts
- Product costs
- Sales strategy

to understand the reasons for negative profitability.

10.4 Monitor High-Sales Products

High-sales products should not be evaluated only on revenue.

Their profitability should also be monitored to ensure that strong sales contribute positively to overall business performance.

11. Conclusion

This project provided practical experience in performing a basic business-oriented data analysis workflow using Microsoft Excel.

The dataset was cleaned and organized before performing analysis. PivotTables were used to calculate important business metrics and compare category and product performance.

The analysis showed that Technology contributed strongly to sales and profit, while Furniture had comparatively lower profitability. The analysis also identified products with negative profit that may require further investigation.

The final Excel dashboard presents the key KPIs, trends, and business insights in a concise and visual format.

Overall, this project helped demonstrate the practical use of **data cleaning, Excel analysis, PivotTables, visualization, dashboard creation, and business insight generation** in a Data Analyst workflow.

12. GitHub Repository

The complete project files have been uploaded to GitHub.

GitHub Repository:

https://github.com/prensi01/ShadowFox_Data_Analyst_Beginner

The repository contains:

- Raw Superstore dataset
- Cleaned Excel analysis file
- Excel dashboard
- README documentation

Thank You

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